

Vision-Based Terrain Classification for Robotic Navigation Using Classical Machine Learning

Project Final Report

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Abstract

This project investigates terrain classification for autonomous robotic navigation using classical computer vision and machine learning techniques. The objective is to classify terrain types from monocular RGB imagery captured in outdoor environments using computationally efficient methods suitable for real-time deployment. Using the Robot Unstructured Ground Driving (RUGD) dataset, image frames are subdivided into overlapping patches and labeled using pixel-level terrain annotations. Histogram of Oriented Gradients (HOG) and Local Binary Patterns (LBP) are extracted from each patch to form engineered feature vectors describing edge and texture characteristics. Multiple machine learning models were evaluated, including Support Vector Machines (SVM), Random Forests, and Logistic Regression. Experimental results showed that Logistic Regression provided the best balance between computational efficiency and classification accuracy, achieving up to 94.11% test accuracy on held-out video sequences. The final system also reconstructs full-frame terrain maps from patch-level predictions. Results demonstrate that classical feature engineering methods remain effective for terrain classification tasks while requiring significantly fewer computational resources than modern deep learning approaches.

1 Introduction

Terrain classification is an important perception task in autonomous robotics. Mobile robots operating in unstructured outdoor environments must distinguish between different surface types in order to navigate safely and efficiently. Terrain properties such as roughness, texture, and material composition directly affect traction, stability, and path planning decisions. A robot capable of understanding terrain composition can make more intelligent navigation decisions and avoid hazardous or inefficient routes.

Modern autonomous systems frequently rely on deep learning for visual perception tasks. While highly effective, deep learning systems typically require extremely large labeled datasets, specialized hardware acceleration, and substantial training time. These requirements can make deployment difficult on embedded robotic platforms with limited computational resources. This project instead explores whether classical computer vision and machine learning methods can achieve useful terrain classification performance while remaining computationally efficient and interpretable.

The dataset used in this project is the Robot Unstructured Ground Driving (RUGD) dataset, which contains RGB video sequences captured from a mobile robot platform alongside pixel-level semantic terrain annotations. These annotations include terrain classes such as dirt, grass, asphalt, and gravel. Rather than performing full semantic segmentation directly, this project reformulates the task as a patch-level terrain classification problem. Each image frame is divided into overlapping patches, and each patch is assigned a terrain label based on the dominant terrain class within the corresponding annotation region.

To represent terrain appearance, this project uses engineered texture and geometry descriptors instead of raw pixel values. Histogram of Oriented Gradients (HOG) captures edge orientation structure and local geometry, while Local Binary Patterns (LBP) capture fine-scale texture information. These feature vectors are used to train traditional machine learning classifiers including Support Vector Machines (SVM), Random Forests, and Logistic Regression models.

The primary goals of this project are:

- Evaluate the effectiveness of classical feature descriptors for terrain classification.
- Compare multiple machine learning models for this task.
- Investigate the tradeoff between classification accuracy and computational efficiency.
- Reconstruct terrain segmentation maps from patch-level predictions.
- Explore whether practical real-time terrain mapping is possible without deep learning.

The final system demonstrates that classical machine learning approaches can produce accurate terrain classifications while maintaining relatively low computational requirements.

2 Related Work

Terrain classification has been widely studied in both robotics and computer vision. Many modern systems use convolutional neural networks (CNNs) for semantic segmentation and scene understanding. CNN-based methods are capable of learning highly expressive hierarchical features directly from image data and often achieve state-of-the-art performance. However, these systems generally require large training datasets, extensive hyperparameter tuning, and powerful GPU hardware.

Prior to the widespread adoption of deep learning, classical feature descriptors were commonly used for visual recognition tasks. These engineered descriptors attempt to capture meaningful image structure through handcrafted representations of texture, gradients, and local spatial relationships.

One of the most influential classical descriptors is Histogram of Oriented Gradients (HOG), introduced by Dalal and Triggs for human detection. HOG computes histograms of local gradient orientations across an image and has proven highly effective for capturing shape and edge information. The descriptor is robust to moderate illumination changes and local geometric variation, making it useful for many recognition problems.

Local Binary Patterns (LBP) are another widely used classical texture descriptor. Rather than focusing on gradient structure, LBP captures local texture by comparing neighboring pixel intensities around each pixel and encoding the resulting binary patterns into histograms. LBP is computationally efficient and effective for representing fine-scale texture variation.

Traditional machine learning algorithms such as Support Vector Machines (SVMs), Logistic Regression, and Random Forests are frequently paired with engineered features. SVMs are particularly effective in high-dimensional feature spaces and can model nonlinear decision boundaries through kernel methods. Random Forests provide efficient ensemble-based classification and are often robust to noisy data. Logistic Regression provides a simpler linear decision model that can still perform well when meaningful features are provided.

This project builds upon these classical methods by combining HOG and LBP descriptors for terrain classification in a robotics context. Rather than competing directly with modern deep learning segmentation systems, this work investigates whether lightweight classical approaches can still provide accurate and computationally practical terrain understanding.

3 Methods

3.1 Dataset and Problem Formulation

This project uses the RUGD dataset, which contains monocular RGB video sequences captured from a robot traversing outdoor environments. Each frame includes a pixel-level semantic terrain annotation image.

The terrain classes used in this project are:

Dirt Grass Asphalt Gravel

Many additional terrain classes exist in the original dataset, but this project focuses only on these four terrain categories. Regions containing other terrain types are treated as unknown terrain.

Instead of directly performing semantic segmentation, the task is reformulated as patch classification. Each image frame is subdivided into overlapping square patches, and each patch is assigned a label corresponding to the dominant terrain within the patch.

Training and testing are performed using image-level splits rather than random patch-level splits. This prevents patches from the same frame from appearing in both training and testing datasets, improving evaluation realism and preventing overly optimistic results.

3.2 Image Processing

All RGB images are converted to grayscale before feature extraction. Since HOG and LBP primarily operate on structural and textural information, color information was not considered necessary for this stage of the project.

Gamma normalization is applied to reduce sensitivity to illumination variation. Pixel values are transformed using gamma correction with a gamma value of 2.2.

Images are then subdivided into overlapping patches. Overlapping patches improve terrain map smoothness and reduce reconstruction artifacts near patch boundaries. Two separate stride values are used:

- Training stride controls the overlap between training patches.
- Testing stride controls the resolution of the reconstructed terrain map.

Reducing testing stride increases terrain map fidelity without affecting model training.

Patches are labeled using the corresponding segmentation maps. A patch is assigned the terrain class occupying the majority of the patch area. During training, patches are discarded unless at least 60% of the patch consists of one of the recognized terrain classes. During testing, a stricter threshold of 90% known terrain is used. Patches failing this threshold are marked as unknown terrain and excluded from quantitative accuracy evaluation.

The system therefore evaluates only patches that contain sufficiently pure examples of the recognized terrain classes.

3.3 Histogram of Oriented Gradients (HOG)

A custom implementation of Histogram of Oriented Gradients was developed using NumPy.

The HOG pipeline proceeds as follows:

1. The image patch is converted into a grayscale intensity array.
2. Symmetric image padding is applied to reduce artificial border gradients.
3. Horizontal and vertical gradients are computed using Sobel filters.
4. Gradient magnitude and orientation are computed at each pixel.
5. The image is partitioned into square cells.
6. For each cell, a histogram of gradient orientations is constructed.
7. Gradient magnitudes contribute weighted votes to neighboring histogram bins using linear interpolation.
8. Neighboring cells are grouped into overlapping blocks.
9. Each block is normalized using L2-Hys normalization.

The final HOG feature vector is formed by concatenating all normalized block histograms into a single high-dimensional representation.

The custom HOG implementation was validated against the scikit-image implementation and achieved strong agreement:

- Correlation: 0.953
- Relative difference: 0.21

This validation provided confidence that the custom implementation was functioning correctly.

3.4 Local Binary Patterns (LBP)

Local Binary Patterns were added later in the project to improve fine-scale texture representation.

LBP operates by comparing each pixel to its neighboring pixels and constructing a binary pattern based on relative intensity comparisons. Histograms of these local binary patterns are then computed over image regions.

While HOG captures larger-scale edge and shape information, LBP captures local texture structure at a finer scale. Combining the two descriptors improved classification performance substantially.

Experiments showed that adding LBP improved Logistic Regression accuracy from approximately 88% to roughly 92–93% while also significantly improving qualitative terrain map reconstruction quality.

3.5 Feature Normalization and Dimensionality Reduction

After feature extraction, HOG and LBP features are concatenated into a single feature vector for each image patch.

Feature standardization is performed using StandardScaler so that each feature dimension has zero mean and unit variance. This prevents large-valued features from dominating the optimization process.

Principal Component Analysis (PCA) was also explored for dimensionality reduction using both scikit-learn PCA and a custom Singular Value Decomposition (SVD) implementation. PCA was intended to reduce redundancy among feature dimensions and improve computational efficiency.

However, experiments showed that retaining 100% of the feature information produced the best visual and quantitative results. PCA slightly reduced terrain map quality and did not significantly improve runtime performance, so dimensionality reduction was ultimately disabled in the final system

3.6 Model Selection

Several machine learning models were evaluated:

- Support Vector Machines (SVM)
- Random Forests
- Logistic Regression

SVMs achieved strong classification accuracy, particularly with RBF kernels, but training and inference times were relatively high. Random Forests provided lower computational cost but struggled with the nonlinear structure of the feature space.

Logistic Regression ultimately provided the best balance between computational efficiency and predictive performance. Despite being a linear model, Logistic Regression performed very well when paired with engineered HOG and LBP features.

Hyperparameter tuning was performed using 5-fold cross validation. The most effective Logistic Regression configuration used:

- Regularization parameter: $C = 0.01$
- No dimensionality reduction
- Standardized feature vectors

The relatively small value of C indicates that strong regularization improved generalization performance, likely due to noise and redundancy in the feature space.

3.7 Terrain Map Reconstruction

After patch-level classification, terrain segmentation maps are reconstructed by projecting patch predictions back onto the original image frame.

Each predicted patch contributes weighted votes to pixels within its spatial region. Overlapping patches produce smoother reconstructions by allowing multiple patch predictions to influence each pixel.

Gaussian-weighted voting strategies were also explored for both patch labeling and terrain reconstruction. These methods weighted the central regions of patches more heavily than boundary regions. While this reduced some boundary artifacts, it also overly smoothed small terrain structures and reduced segmentation detail. Uniform voting ultimately produced better qualitative results.

Final terrain maps are reconstructed by selecting the terrain class with the maximum accumulated vote at each pixel.

4 Experimental Results

4.1 Experimental Setup

Experiments were conducted on an AMD Ryzen 9 5900HS processor with 16 GB RAM on an ASUS ROG Zephyrus G15 laptop. GPU acceleration was not used.

Multithreading was implemented for HOG and LBP feature extraction using all available CPU cores. This significantly reduced preprocessing time and enabled faster experimentation.

Final evaluation metrics were computed using held-out testing videos rather than cross-validation folds.

4.2 Classification Performance

The following results summarize the best-performing experiments.

Model	Training Videos	Testing Videos	Patch Size	Train Stride	Test Accuracy	Processing Time
Logistic Regression	trail-9 (59 frames)	trail-10 (49 frames)	81×64	16	94.11%	N/A
Logistic Regression	trail-9 (59 frames)	trail-10 (49 frames)	64×64	16	91.11%	N/A
Logistic Regression	trail:subset (224/693 frames)	trail:subset (469/693 frames)	64×64	16	88.50%	0.531 sec/frame

Table 1: Final terrain classification performance across multiple experimental configurations.

Additional models achieved:

- RBF SVM: approximately 93.5% accuracy
- Random Forest: approximately 83% accuracy

The highest accuracies were achieved on the trail-9 and trail-10 split, which primarily contained grass and gravel terrain classes. More diverse datasets containing all four terrain classes proved substantially more difficult.

4.3 Effect of Patch Size

Patch size had a noticeable effect on both accuracy and terrain map quality.

Larger patches provided more contextual information and generally improved classification accuracy. The 81×64 patch configuration achieved the highest overall accuracy of 94.11

However, larger patches reduce terrain map precision because each prediction covers a larger spatial region. Smaller patches produce more detailed reconstructions but may lack sufficient contextual information for accurate classification.

This tradeoff between contextual accuracy and segmentation fidelity was one of the key observations of the project.

4.4 Effect of Overlapping Patches

Overlapping patches significantly improved reconstruction quality.

Training overlap increases the number of training examples but also introduces redundancy into the dataset. Testing overlap primarily affects reconstruction fidelity rather than model quality.

Reducing testing stride from larger values down to 8 pixels produced smoother and more spatially detailed terrain maps. However, smaller strides also increased computational cost because more patches needed to be processed.

RBFSVM models produced smoother reconstructions at larger testing strides than Logistic Regression, although the increased computational cost reduced their practicality for real-time use.

4.5 Effect of Feature Engineering

The combination of HOG and LBP descriptors proved substantially more effective than HOG alone.

HOG-only models achieved approximately 88% accuracy under similar conditions. Adding LBP increased accuracy to roughly 92–93% and dramatically improved reconstruction quality.

This suggests that HOG and LBP capture complementary information:

HOG captures larger geometric and edge structures. LBP captures finer local texture variation.

Together, the descriptors produce a richer terrain representation than either method alone.

4.6 Runtime Performance

The final Logistic Regression pipeline achieved approximately 0.531 seconds per testing frame, corresponding to roughly 2 frames per second.

This timing included:

- Image loading
- Patch extraction
- HOG extraction
- LBP extraction
- Feature normalization
- Model inference
- Terrain reconstruction

Although not yet suitable for high-speed robotics applications, the system demonstrates that practical terrain segmentation can be achieved using classical machine learning without GPU acceleration.

5 Conclusion

This project implemented a terrain classification system for autonomous robotic navigation using classical computer vision and machine learning techniques. Using HOG and LBP feature descriptors combined with traditional classifiers, the system successfully classified terrain patches and reconstructed full-frame terrain maps from monocular RGB imagery.

Experimental results demonstrated that classical feature engineering approaches remain highly effective for terrain classification tasks. Logistic Regression provided the best balance between computational efficiency and classification accuracy, achieving up to 94.11% accuracy on held-out testing videos. The addition of LBP features significantly improved both quantitative accuracy and qualitative reconstruction quality.

Several important tradeoffs were observed throughout the project, including the relationship between patch size, reconstruction fidelity, and computational cost. Overlapping patches improved terrain map quality, while dimensionality reduction through PCA provided little benefit.

Future work could focus on implementing GPU acceleration, exploring stochastic gradient descent classifiers, investigating superpixel-based segmentation instead of square patches, and developing methods for automatic unknown-terrain detection using prediction confidence thresholds. Additional work could also evaluate whether LBP alone is sufficient for accurate terrain classification or whether more advanced texture descriptors could further improve performance.

Overall, this project demonstrates that lightweight classical machine learning methods can provide accurate and computationally practical terrain understanding for robotic systems.

6 References

RUGD Dataset:

RUGD Dataset

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